Data Structures Lab (CL2001)

Date: October 16th 2026

Section: C

Sessional Exam

Total Time (Hrs):	**2 hrs**
Total Marks:	**20**
Total Questions:	**2**

Roll No Section Student Signature

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Attempt all the questions.

CLO #: 1

Q1: [11 Marks]

A rat is placed at the top left cell of a maze given as a 2D matrix. A cell with **0** is open and the rat can move into it, a cell with **1** is blocked and the rat must stop there. The rat has to reach the bottom right cell. It can move Down (D), Left (L), Right (R) and Up (U), and it can not visit the same cell twice in a single path.

Write a **recursive program** to print **all possible paths** from the start to the end as a string of moves. Try the moves in the order D, L, R, U. Your path search must use recursion with **backtracking**. Implement the maze as a class **Maze** that keeps the grid, a visited array, the row and column count and the current path.

0	0	0	0
1	0	1	0
0	0	0	0

Output:

```
RDDRR
RRRDD
Total paths: 2
```

Green cell: start (0,0). Pink cell: end (2,3). Grey cells: blocked (1).

a) [2 Marks] Function that checks whether the rat can step into a cell (inside the maze, not blocked and not already visited in the current path)

```
bool isSafe(int r, int c)
```

b) [6 Marks] Recursive function with backtracking that prints every path from (r, c) to the bottom right cell. The current path is built move by move and printed when the end is reached.

```
void solve(int r, int c, int len)
```

c) [3 Marks] Function that starts the search from the top left cell and returns the number of paths found. If the start or the end cell is blocked it must return 0. Print "Total paths: N" or "No path found" at the end.

```
int findPaths()
```

CLO #: 2

Q2: [9 Marks]

An array stores the temperature of n consecutive days. For every day, find how many days you have to wait until a **warmer** temperature comes. If there is no warmer day in the future, the answer for that day is 0. Put the answers **back in the same array**.

Input:	`73 74 75 71 69 72 76 73`
Output:	`1 1 4 2 1 1 0 0`

a) [3 Marks] Implement a **stack using a linked list**. Make a class Node and a class Stack with push, pop, top, isEmpty and a destructor that frees the remaining nodes. Do not use the STL stack.

b) [5 Marks] Use your stack to solve the problem in a **single traversal** of the array.

```
void dailyTemperatures(int temps[], int n)
```

c) [1 Mark] Create a Menu for all questions.

Rules: Only the iostream library is allowed. No vectors, hashmaps or any other STL container. Use classes instead of structs. Write your own stack, do not use the STL one.